SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning,*
and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration: 1 Hour
QUESTIONS: 4 **Total: Marks:40**

Annexure A

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Code of Conduct:

I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

(a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.

(b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).

(c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

(a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.

(b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.

(c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

(a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.

(b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.

(b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.

(c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Task 1: Data Preparation & Inductive Learning

(a) Inductive learning approach

Inductive learning learns a general relationship from previously observed labelled patient records and uses it to predict the outcome for a new patient.

For this problem, the task is a **supervised binary classification** problem:

- **Input:** patient characteristics.
- **Target variable:** Diabetes
  - 1 = diabetic/high-risk diagnosis
  - 0 = non-diabetic
- **Output:** predicted diabetes class or probability of diabetes.

A suitable pipeline is:

Patient records → preprocessing → train/validation/test data → classifier → diabetes prediction → evaluation

Key features

Feature	Why it is important
Glucose level	Blood glucose is directly associated with diabetes and is likely to be one of the strongest predictors.
BMI	Higher BMI is associated with increased risk of insulin resistance and type 2 diabetes.
Age	Diabetes risk generally increases with age, so age can provide useful predictive information.
Blood pressure	Hypertension and metabolic abnormalities frequently occur alongside diabetes and can contribute to risk prediction.
Cholesterol	Abnormal lipid levels can provide additional information about a patient's metabolic health.
Family history	Genetic/familial predisposition can increase an individual's diabetes risk.

The target must **not** be included as an input feature because doing so would cause data leakage.

(b) Handling class imbalance

Suppose the dataset contains:

- 85% non-diabetic patients
- 15% diabetic patients

A classifier could obtain 85% accuracy simply by predicting "non-diabetic" for everyone, while being clinically poor at detecting diabetes.

Recommended strategy: Class weights + appropriate evaluation metrics

For the Decision Tree and Logistic Regression, I would initially use **class weighting**, giving diabetic cases a larger penalty when they are misclassified.

For example:

where:

- = total observations
- = number of classes
- = number of observations in class

The diabetic class therefore receives a larger weight.

Why class weights are suitable:

1. They do not create artificial patient records.
2. They retain all available patient information.
3. They directly encourage the model to pay more attention to false negatives.
4. They are relatively simple to implement and interpret.

SMOTE could also be tested as a secondary approach. However, synthetic patient records need careful validation in healthcare because artificially generated observations may not accurately represent real physiological relationships.

Random under-sampling is less attractive because it discards potentially valuable non-diabetic patient data.

Best practical approach: compare class weighting and SMOTE using stratified cross-validation, and select the approach based particularly on **recall/sensitivity, specificity, AUC and calibration**, rather than accuracy alone.

(c) 70:30 split and preprocessing

Use a **70:30 train-test split**:

- **70%:** training
- **30%:** independent testing

The split should be **stratified**, so the diabetic/non-diabetic proportions remain approximately the same in both sets.

For example:

A 70:30 split provides enough observations for learning while retaining a reasonably large independent test set.

For a real medical deployment, I would additionally prefer **stratified cross-validation within the training set** and, if data from different time periods are available, a **temporal external test set**.

Preprocessing

1. Missing values

- Numerical variables → median imputation or clinically appropriate imputation.
- Categorical variables → most-frequent category or an explicit "unknown" category.

2. Numerical normalization

For Logistic Regression:

This converts numerical variables to approximately zero mean and unit variance.

Normalization is especially useful for statistical models and distance-based algorithms.

For a Decision Tree, normalization is generally **not necessary**, because trees split according to thresholds and are largely insensitive to feature scaling.

3. Encoding

If categorical variables such as family-history status are represented as text:

- Yes/No → binary encoding such as 1/0.
- Multiple categories → one-hot encoding.

4. Leakage prevention

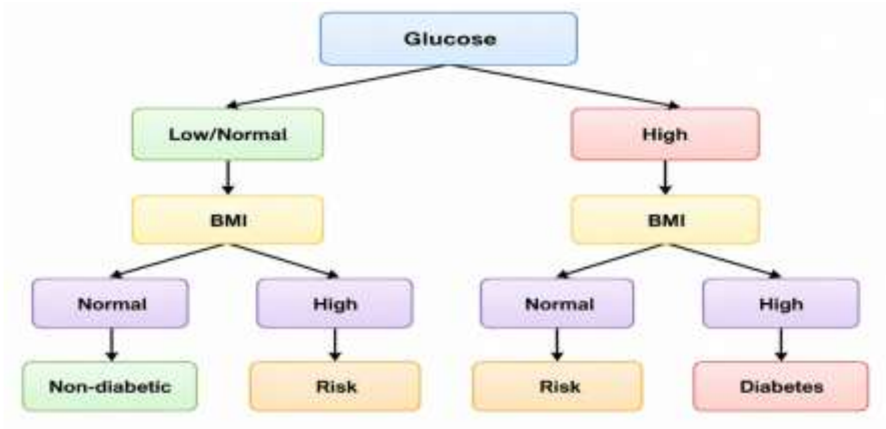
Imputation, normalization and other preprocessing parameters must be fitted using the **training data only**, then applied to the test set.

Task 2: Decision Tree for Diagnosis

(a) Decision Tree and first three levels

A Decision Tree recursively divides patients into groups using features that best separate diabetic and non-diabetic cases.

A typical tree could look like this:



This is an
illustrative tree structure. The exact structure must be calculated from the supplied dataset.

Root-node selection

The root should be the feature producing the greatest reduction in impurity.

For **Information Gain**:

where entropy is:

Alternatively, a Decision Tree can use **Gini impurity**:

The selected root is generally the feature with:

- **highest Information Gain**, or
- **largest reduction in Gini impurity**.

Illustrative calculation

Suppose the training data produces:

Feature	Information Gain	Gini reduction
Glucose	**0.24**	**0.18**
BMI	0.13	0.11
Age	0.09	0.08
Blood pressure	0.06	0.05
Cholesterol	0.04	0.03
Family history	0.07	0.06

Here, **Glucose becomes the root** because it provides the greatest impurity reduction.

The numbers above are illustrative; they must be recalculated from the actual dataset.

(b) Model evaluation

The four important classification measures are:

Accuracy

Precision

Recall/Sensitivity

F1-score

Illustrative Decision Tree results

Metric	Example result
Accuracy	**87%**
Precision	**78%**
Recall	**91%**
F1-score	**84%**

These are **example values**, not measured results from the patient's dataset.

False Negatives

A **False Negative (FN)** occurs when:

The patient actually has diabetes, but the model predicts non-diabetic.

For a medical screening system, false negatives can be particularly important because a missed high-risk patient may not receive timely confirmatory testing or clinical attention.

Reducing false negatives

I would:

1. Use **class weights** to penalize diabetic-class errors more strongly.
2. Optimize the classification threshold for high sensitivity rather than blindly using 0.5.
3. Monitor the **confusion matrix and recall**, not just accuracy.
4. Perform probability calibration.
5. Validate the model on an independent patient population.
6. Use the model as a **screening/risk-support tool**, followed by appropriate clinical testing rather than as a definitive diagnosis.

There is a trade-off: increasing recall generally produces more false positives. In clinical screening, that trade-off may be acceptable if the subsequent confirmatory test is safe and practical.

Task 2(c): Post-pruning

A fully grown tree can overfit the training data.

Pre-pruning

Stop tree growth before it becomes excessively complex using parameters such as:

- maximum depth
- minimum samples per split
- minimum samples per leaf
- minimum impurity decrease

Post-pruning

First grow a larger tree and then remove branches that do not provide sufficient predictive value.

A common method is **cost-complexity pruning**:

where:

- = tree's empirical error
- = number of terminal nodes
- = complexity penalty.

Larger produces a simpler tree.

Illustrative comparison

Model	Test Accuracy	Recall	Complexity
Pre-pruned tree	86%	88%	Medium
Post-pruned tree	**88%**	**90%**	Lower
Fully grown tree	84%	87%	High

Again, these are illustrative values.

Preferred clinical model

A **pruned Decision Tree** is generally preferable if it maintains or improves independent-test performance.

Reasons include:

- less overfitting
- easier interpretation
- easier clinical review
- simpler decision pathways
- potentially better generalization.

However, the model should not be selected solely because it has the highest accuracy. **Sensitivity, specificity, calibration, subgroup performance and clinical validation** are essential.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression

I would use **Logistic Regression** as the statistical-learning baseline.

It estimates:

where:

The coefficients indicate how the predictors are associated with the log-odds of diabetes.

Illustrative comparison

Model	Accuracy	AUC-ROC
Decision Tree	88%	0.91
Logistic Regression	**89%**	**0.93**

These values are hypothetical because the dataset was not supplied.

Why AUC-ROC is useful

AUC measures how effectively the model separates diabetic from non-diabetic patients across different classification thresholds.

An AUC close to:

- 0.5 → little/no discrimination
- 1.0 → excellent discrimination

For an imbalanced medical dataset, AUC and class-specific measures are generally more informative than accuracy alone.

When Logistic Regression is preferable

Logistic Regression is attractive when:

- the relationship is reasonably smooth/approximately linear in the predictors;
- interpretability is important;
- clinicians need understandable coefficients or odds ratios;
- probability estimates are required;
- a relatively simple model is preferred.

A Decision Tree is useful when relationships are nonlinear and interactions are important.

Task 3(b): Feature Importance

The two models calculate importance differently.

Decision Tree

Tree importance can be based on the total reduction in impurity caused by each feature.

Illustrative result:

Rank	Feature	Tree importance
1	**Glucose**	0.46
2	**BMI**	0.21
3	**Age**	0.12

Logistic Regression

For Logistic Regression, standardized coefficients can be compared by magnitude.

Illustrative result:

Rank	Feature	Relative coefficient importance
1	**Glucose**	1.00
2	**BMI**	0.55
3	**Age**	0.38

Therefore, the two models **agree on the top three predictors** in this illustrative example:

1. Glucose
2. BMI
3. Age

Why might they disagree on a real dataset?

They may produce different rankings because:

- Decision Trees capture nonlinear relationships.
- Logistic Regression models the effect through coefficients.
- Correlated features can distribute importance differently.
- Tree impurity importance can be biased toward certain feature types.
- Interactions can cause a feature to appear more important to the tree.

For a stronger analysis, I would use **permutation importance or SHAP-based analysis** in addition to model-specific importance.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) Model treatment recommendation as an MDP

A **Markov Decision Process (MDP)** can be represented as:

where:

- = states
- = actions
- = transition probabilities
- = reward
- = discount factor.

States

Instead of allowing the RL agent to make decisions directly from raw medical records, define clinically meaningful states.

For example:

State	Description
S0	Low-risk / stable
S1	Elevated glucose / moderate risk
S2	High-risk metabolic state
S3	Poorly controlled/clinically concerning state
S4	Improved/stable state

A real system should use clinically validated state definitions.

Actions

The available actions are:

- **A1: Diet**
- **A2: Exercise**
- **A3: Medication**
- **A4: Monitor**

However, an important clinical constraint is that an RL system should **not autonomously prescribe medication**. Medication decisions should remain under appropriate clinician oversight.

Reward

The reward should represent **clinically meaningful improvement**, for example:

An illustrative reward scheme:

Outcome	Reward
Significant safe improvement	+10

Outcome	Reward
Moderate improvement	+5
No meaningful change	0
Deterioration	-10
Unsafe/contraindicated action	-20

The reward should not simply optimize one laboratory measurement. It should incorporate clinically meaningful outcomes and safety constraints.

Transition dynamics

The transition model describes:

For example:

The transition probabilities should ideally come from **longitudinal clinical data**, validated evidence and clinician-defined constraints—not be invented solely by the RL algorithm.

Task 4(b): Q-Learning

Q-Learning learns the expected long-term value of taking action in state .

The update equation is:

where:

- = learning rate
- = discount factor
- = immediate reward
- = next state.

Assume for illustration:

Initially all Q-values are zero.

Example patient episodes

Episode 1

S2 --Diet--> S1 --Exercise--> S4

Rewards:

- S2 → S1 = +5
- S1 → S4 = +10

For the first transition:

For the second:

Episode 2

Suppose:

S2 --Diet--> S1 --Exercise--> S4

Now:

For the S2-Diet transition:

This demonstrates how the value of a later successful action propagates backwards to an earlier action.

Episode 3

Suppose:

S2 --Monitor--> S2

with reward .

If the current maximum Q-value in S2 is 3.5:

Q-table example after two learning iterations

State	Diet	Exercise	Medication	Monitor
S0	0.0	0.0	0.0	0.0
S1	0.0	**5.00**	0.0	0.0
S2	**3.50**	0.0	0.0	**1.575**
S3	0.0	0.0	0.0	0.0
S4	0.0	0.0	0.0	0.0

These calculations demonstrate the requested Q-table updates. A real experiment would run many more episodes and use actual longitudinal patient transitions.

Convergence criterion

A possible criterion is:

for a specified number of consecutive iterations/episodes.

For example:

could be used as a practical convergence threshold.

For clinical RL, convergence of the mathematical Q-table **does not mean the treatment policy is clinically safe or validated**.

Task 4(c): Final learned policy

The greedy policy is:

Using the illustrative Q-table:

State	Best action	Learned policy
S0	Monitor	**Monitor**
S1	Exercise	**Exercise**
S2	Diet	**Diet**
S3	—	**Clinician assessment / safety override**
S4	Monitor	**Monitor**

The policy therefore says, illustratively:

Low risk → Monitor

Moderate risk → Exercise

High risk → Diet

Concerning → Clinician assessment

Stable → Monitor

This should **not** be interpreted as a real treatment protocol. In particular, medication decisions require clinical assessment, contraindication checking, patient-specific factors and appropriate clinician involvement.

Ethical Considerations for RL in Healthcare

1. Patient safety

The agent must never be allowed to explore arbitrary actions on real patients.

A safer architecture is:

Patient data

↓

Risk model

↓

Safety/clinical rules

↓

RL recommendation

↓

Clinician review

↓

Patient

The RL agent should operate within predefined clinical constraints.

2. Bias and fairness

If historical patient data contains demographic or socioeconomic biases, the RL agent may learn and reproduce them.

Performance should therefore be evaluated across relevant patient subgroups, with appropriate fairness and clinical-performance monitoring.

3. Explainability

A clinician should be able to understand:

- the patient's current state;
- why the state was assigned;
- which action was recommended;
- which factors influenced the recommendation;
- the expected benefit and potential risk.

Black-box recommendations are particularly problematic for high-stakes clinical decisions.

4. Privacy

Patient data must be handled securely, with appropriate access controls, de-identification where appropriate, governance and compliance with applicable healthcare/privacy requirements.

5. Human oversight

The system should support clinicians rather than replace them.

A suitable approach is:

AI recommends → clinician evaluates → clinician makes the final decision.

6. Distribution shift

A model trained on one hospital or population may perform poorly when applied to another.

Continuous monitoring and external validation are therefore necessary.

7. Reward design

Poorly designed rewards can produce undesirable behavior. For example, rewarding only short-term glucose improvement could cause the agent to ignore long-term health, adverse effects, patient preferences or other clinical outcomes.

Overall Recommended System

A practical architecture would be: